

Option A – Alternating Currents

6. (a) The sinusoidal emf produced by a rotating coil in a magnetic field has a **peak** emf of 18.0 V and a frequency of 70.0 Hz. The coil is connected to the y-input of an oscilloscope.
- (i) The output of the rotating coil is used to power a small lamp of known resistance. Explain how you would calculate the rms input power to the lamp from the known resistance and **peak** emf. [2]

.....

.....

.....

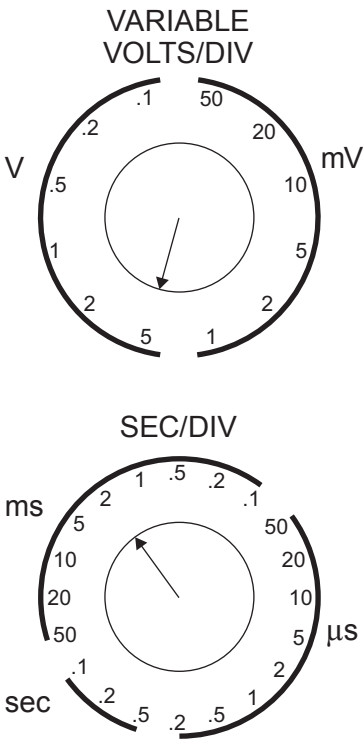
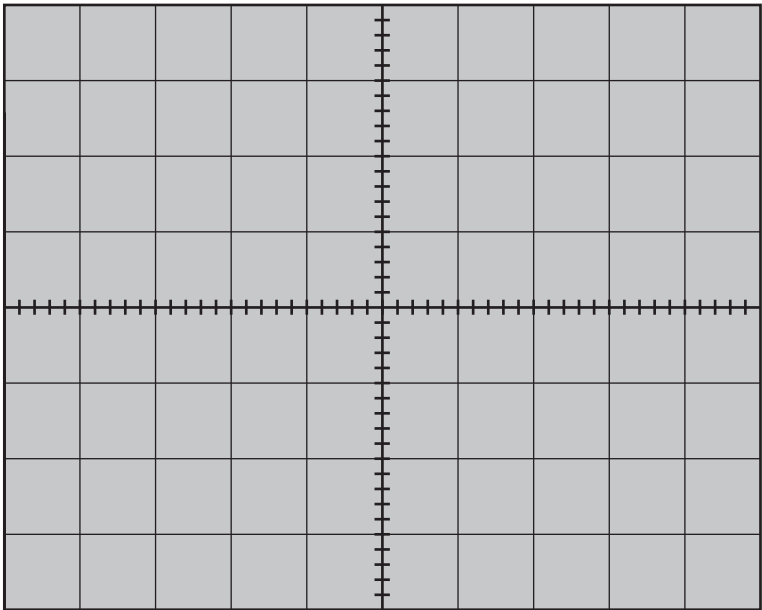
.....

.....

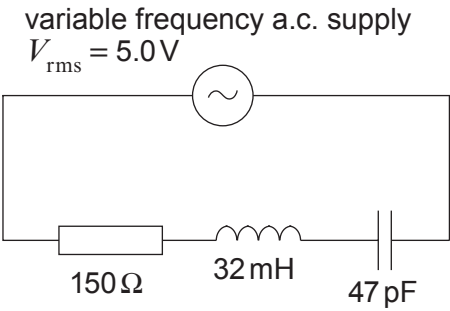
.....

- (ii) Draw a typical trace that might be seen on the oscilloscope screen with the oscilloscope settings shown. [5]

Space for calculations.



(b) An *LCR* circuit is shown below.



(i) Explain why the resonance frequency of the circuit occurs when: [3]

$$\omega L = \frac{1}{\omega C}$$

.....

.....

.....

.....

.....

.....

(ii) Calculate the resonance frequency of the circuit. [2]

.....

.....

.....

.....

(iii) Calculate the rms current when the frequency of the supply is 324 kHz. [3]

.....

.....

.....

.....

.....

.....

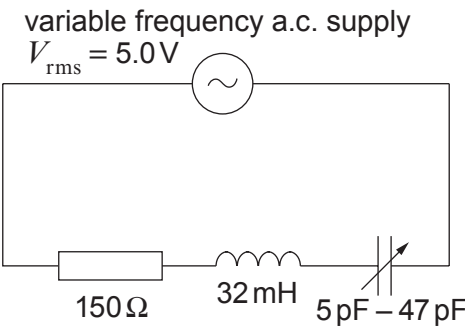
.....



Examiner
only

- (iv) A student claims that the following circuit cannot have a peak pd above 1.5 kV across the capacitor. Investigate whether or not he is correct. [5]

(It may be useful to note that $\frac{\omega_0 L}{R} = \frac{1}{R} \sqrt{\frac{L}{C}}$.)



.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

20

